

VLSI EDA Toolkit Development using C

Industry-Style Internship Project Handbook

Project Duration

8 Weeks

Difficulty

Intermediate to Advanced

Objective

Develop a mini Electronic Design Automation toolkit in C for parsing Verilog/SPICE netlists, timing analysis, design rule checking and logic optimization.

Technology Stack

C, GCC, Makefile, GDB, Valgrind, Verilog/SPICE Netlists, GTKWave, Icarus Verilog, Yosys, optional Cadence validation.

Modules

Netlist Parser, Circuit Graph Generator, Gate Connectivity Analyzer, Mini STA, Power Estimator, Logic Optimizer, DRC, Report Generator.

Week 1

C programming refresher, parser skeleton.

Week 2

Verilog parser and AST.

Week 3

Circuit graph, DFS/BFS, topological sort.

Week 4

Static timing analysis.

Week 5

Design Rule Checker.

Week 6

Logic optimization.

Week 7

Power estimation and CSV reports.

Week 8

Integration, documentation and presentation.

Deliverables

Source code, Makefile, parser library, reports, technical documentation and presentation.

Evaluation

C Programming 25%, Parser 20%, Graph Algorithms 15%, STA 15%, DRC 10%, Optimization 10%, Documentation 5%.

Learning Outcomes

EDA software development, parsing, graph algorithms, timing analysis, DRC, optimization and debugging.